Interoperability Conformance Specification: hybridPrinterWithReflectance

Contents

Foreword

This document has been prepared following the ICC Intellectual Property Policy. This policy is based on the ITU-T/ITU-R/ISO/IEC Guidelines for Implementation of the Common Patent Policy (23 April 2012), with interpretations and clarifications to make it specific to ICC. A Patent Statement and Licensing Declaration form is available.

ICC Interoperability Conformance Specifications, of which this document is an example, may be submitted to the competent ISO Technical Committee for consideration and development as an ISO document. If so, this foreword is to be replaced by the appropriate wording supplied by ISO.

Introduction

ISO 20677-1 defines specifications that provide a platform for defining extended (iccMAX) colour management profiles and systems for various colour workflow domains. It provides a platform for which domain specific specifications can be defined that make use of iccMAX extensions to the existing cross-platform profile format of ISO 15076-1 (ICC) with greater flexibility for defining colour transforms and profile connection spaces to meet needs outside the scope of ICC profiles. However, not all systems provide support for iccMAX profiles thus limiting the ability to take advantage of extended capabilities at strategic points in the colour management ecosystem. Using a hybrid ICC profile that has an embedded iccMAX profile as a tag allows for compatibility with ICC systems that do not support iccMAX functionality while providing systems in the colour management ecosystem with iccMAX extensions to take advantage of extended colour management capabilities enabled by iccMAX profiles.

Requirements specifying restrictions to iccMAX that apply to a particular workflow are defined in workflow domain specifications known as Interoperability Conformance Specifications, of which this document is one example. Additionally, for some domain specific workflows it is envisioned that workflows will connect both to ICC profiles, iccMAX profiles, and hybrid profiles containing both ICC and iccMAX content.

An Interoperability Conformance Specification (ICS) is approved and registered by the International Color Consortium (ICC). It defines minimum structural and operational requirements for writing and reading iccMAX profiles in order to address a specific problem and/or functionality that cannot readily be handled using the ICC profile format defined by ISO 15076-1. An ICS document essentially defines restrictions to iccMAX content in profiles for specific use cases.

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1 Scope

This specification defines workflow requirements and restrictions for hybrid ICC profiles for printing that contain an iccMAX sub-profile as a tag for the purpose of providing connection with a spectrally-based Profile Connection Space.

The particular sub-set of the tags in the iccMAX sub-profle defined in ISO 20677-1 that are required to be present is defined, together with any optional tags that are permitted. The connections between profiles are described, and the processing elements that the CMM is required to support are identified.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 15076-1:20##, *Image technology colour management — Architecture, profile format, and data structure*

NOTE: The most recent version of the ICC specification is available on the ICC web site [2].

ISO 20677-1:20##, *Image technology colour management — Extensions to architecture, profile format, and data structure: iccMAX*

NOTE: The most recent version of the iccMAX specification is available on the ICC web site [2].

3 Terms and definitions

All terms and definitions relevant to this document are provided in ISO 15076-1 and ISO 20677.

4 Use case

4.1 Domain

Profiles and workflows conforming to this ICS shall apply to the domain of colour management of printing processes including colour reproduction as well as previewing/proofing printed output.

4.2 Intended use

The intended use of this specification is to define workflows in which either colour data is converted to ink amounts for printing or ink amounts for printing are converted to another colour data representations for proofing, display, or other re-purposing aims. The ICC profile portion of a hybrid printer profile (defined by ISO 15076-1) provides transforms between ink amounts and a colorimetric representation. The iccMAX portion of a hybrid printer profile (defined by ISO 20677-1) provides transforms between ink amounts and a spectral colour representation (where subsequent conversion to colorimetry can be achieved by applying the illuminant and observer specified in the spectralViewingConditionsTag).

4.3 Restrictions

ISO 15076-1 provides full details for an ICC printer profile. This document defines a set of restrictions which apply to an iccMAX sub-profile when it is embedded into an ICC printer profile, as a component of a hybrid profile created for the specific use cases described above. Such restrictions include the sub-set of tags from ISO 20677-1 which are permitted in such an iccMAX sub-profile conforming to this document.

5 Workflow

5.1 Hybrid sub-profile sub-classes

The profile class of a hybrid iccMAX sub-profile shall match the class of the containing ICC profile.

A supporting CMM shall support the profile classes and sub-profile sub-classes identified in Table 1 for conformance with this ICS. All profile classes are defined in ISO 15076-1 and ISO 20677-1.

Table 1. Profile classes and subclasses defined by this ICS

Profile	Profile class	Sub-profile sub-class identifier	Class signature	Sub-class signature
P	printer	spectralRefectance	'prn '	'sref'

5.2 Connection Scenarios

5.2.1 General

The ICC profile portion of a hybrid printer profile is intended to be used as either a source profile or a destination profile by any system that supports ICC printer class profiles.

The 'sref' spectralReflectance sub-profile in a hybrid printer profile connects *x* input channels (where *x* corresponds to the number of channels of the Data colour space field of the header) to a spectral Profile Connection Space containing *n* spectral channels (where *n* corresponds to the number of spectral PCS channels defined in the header). It can be used as either a source profile, or a spectral reproduction target profile using a search based CMM.

5.2.2 Profiles for ICS workflow scenarios

Table 2 provides overview information about additional profiles that are used to describe several profile connection scenarios conforming to this ICS.

Table 2. Additional profiles referenced in workflow scenarios defined by this ICS

Profile	Description

C	Profile conforming to ISO 20677-1 or ISO 15076-1 containing (a) colorimetric transform(s) with CMM configured to use a colorimetric transform type. If a separate ICS specification is associated with this profile it should be followed in addition to the connection requirements of this document.
S	Profile conforming to ISO 20677-1 containing (a) spectral reflectance transform(s) with the CMM configured to use a spectral transform type. If a separate ICS specification is associated with this profile it should be followed in addition to the connection requirements of this document.
L	Profile conforming to ISO 20677-1 or ISO 15076-1 containing CIELAB device encoding using (a) colorimetric transform(s) with CMM configured to use a colorimetric transform type
1 2 N	Profiles conforming to ISO 20677-1 containing Profile Connection Condition tags (spectralViewingConditionsTag, customToStandardPccTag, standardToCustomPccTag)

5.2.3 Workflow Scenarios

5.2.3.1 General

The following sections define scenarios that conform to this ICS.

Note: systems that only support ICC profiles will only work with scenarios 5.2.3.2 and 5.2.3.3.

5.2.3.2 Scenario 1 – Connecting a source profile to a hybrid printer profile as destination with a colorimetric PCS

Profile Sequence	Profile Setup
C → PCS → P	C: Rendering Intent: Any Transform Type: Colorimetric PCS Override: None Use iccMAX Sub-profile: - P: Rendering Intent: Any Transform Type: Colorimetric PCS Override: None Use iccMAX Sub-profile: No

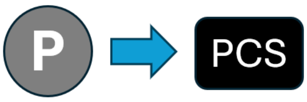
This workflow may be referred to as the "printing" workflow as it uses the base profile transforms in the hybrid printer with reflectance profile to generate data ready for printing. The iccMAX sub-profile is not used for profile P, the base profile is used instead. This is the default behaviour of systems that do not support iccMAX profiles when this profile is used as a destination profile.

5.2.3.3 Scenario 2 – Connecting a hybrid printer profile as source to a destination profile with a colorimetric PCS

Profile Sequence	Profile Setup
	 Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No  Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -

This workflow may be referred to as a "proofing" or "simulation" workflow as it uses the base profile transforms in the hybrid printer with reflectance profile to convert print ready data to colorimetry for display or output. The iccMAX sub-profile is not used for profile P, the base profile is used instead. This is the default behaviour of systems that do not support iccMAX profiles when this profile is used as a source profile.

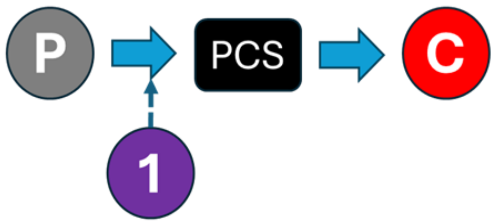
5.2.3.4 Scenario 3 – Getting Spectral PCS data directly from a hybrid printer profile

Profile Sequence	Profile Setup
	 Rendering Intent: Any Transform Type: Spectral PCC Override: None Use iccMAX Sub-profile: Yes

This workflow may be referred to as a spectral "access" workflow as it only uses the iccMAX sub-profile transforms in the hybrid printer with reflectance profile to get spectral reflectance PCS data directly. No spectral PCS operations are performed. Systems that do not support iccMAX profiles do not support this workflow. This is a limited access workflow that doesn't require iccMAX PCS operation, and can be implemented with limited extension of MPE processing defined in ISO 15076-1.

5.2.3.5 Scenario 4 – Connecting a hybrid printer profile as source using a PCC override to a destination profile with an alternate colorimetric PCS

Profile Sequence	Profile Setup



P	Rendering Intent:	Any
	Transform Type:	Spectral
	PCC Override:	1
	Use iccMAX Sub-profile:	Yes
C	Rendering Intent:	Any
	Transform Type:	Colorimetric
	PCC Override:	None
	Use iccMAX Sub-profile:	-

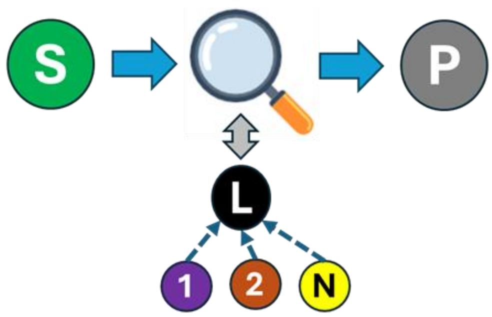
This workflow may be referred to as a "proofing" or "simulation" workflow as it uses the iccMAX sub-profile transforms in the hybrid printer with reflectance profile to convert print ready data to colorimetry for display or output. An iccMAX sub-profile transform is used to get spectral reflectance PCS data which is converted to colorimetry in conjunction with Profile Connection Condition tags to get the colorimetry which is passed to the destination profile. Systems that do not support iccMAX profiles do not support this workflow.

5.2.3.6 Scenario 5 – Connecting a hybrid printer profile as source to a destination profile with a spectral PCS

Profile Sequence	Profile Setup		
	P	Rendering Intent:	Any
		Transform Type:	Spectral
		PCC Override:	None
		Use iccMAX Sub-profile:	Yes
	S	Rendering Intent:	Any
		Transform Type:	Spectral
		PCC Override:	None
		Use iccMAX Sub-profile:	-

This workflow may be referred to as a spectral "sampling" or "conversion" workflow as it uses the iccMAX sub-profile transforms in the hybrid printer with reflectance profile convert print ready data to spectral reflectance which is passed to the destination profile. Spectral PCS operations may be utilized as outlined in ISO 20677-1. Systems that do not support iccMAX profiles do not support this workflow.

5.2.3.7 Scenario 6 – Connecting source profile with a spectral PCS to a hybrid printer profile as destination

Profile Sequence	Profile Setup		
	S	Rendering Intent:	Any
		Transform Type:	Spectral
		PCC Override:	None
		Use iccMAX Sub-profile:	-
	L	Rendering Intent:	Any
		Transform Type:	Colorimetric
		PCC Override:	1 2 … N
		Use iccMAX Sub-profile:	-
	P	Rendering Intent:	Any
		Transform Type:	Spectral
		PCC Override:	
		Use iccMAX Sub-profile:	Yes

This workflow may be referred to as a spectral "reproduction" workflow. This workflow requires special CMM processing as it uses the source iccMAX sub-profile transforms in the hybrid printer with reflectance to perform a search that results in spectral reflectances that have a minimized cost relationship with the spectral reflectances that are passed into the source profile. The minimized cost relationship may for example utilize an index of metamerism computation which involves first establishing desired colorimetry using an interim Lab colour space profile along with two or more Profile Connection Conditions. Then print data values are applied to the hybrid printer with reflectance profile to get reflectance data that is also converted to colorimetry with the same interim Lab colour space profile and PCC profiles. The PCC based colorimetry of the target reflectance is compared to weighted PCC based colorimetry for the print data to adjust print data values to minimize the compared differences. A search is performed until minimal differences are achieved, a minima is found, or a limit of search retries is reached. Spectral PCS operations may be utilized as outlined in ISO 20677-1. Systems that do not support iccMAX profiles do not support this workflow.

Note: The iccApplySearch tool of the iccDEV project provides an example of performing such a search with a printer Hybrid Profile.

6 Sub-class Profile Requirements

6.1 General

Requirements for hybrid iccMAX profiles conforming to this ICS document are listed here. ICC v4 profiles shown in Section 5 above shall conform to ISO 15076-1.

6.2 Requirements

The requirements for a hybrid printer profile with reflectance are in two parts.

The encoding of the base profile shall be as defined in ISO 15076-1 as a printer class profile.

An extension iccMAX sub-profile shall also be encoded in this base (encapsulating) profile as defined in the embedding annex in ISO 20677-1. This involves adding an embeddedV5ProfileTag (signature 'ICC5') to the base profile, with the tag encoded as an embeddedProfileType (signature 'ICCP') with the sub-profile encoded as defined in ISO 20677-1.

The encoding of the profile header of the encapsulated sub-profile in the embeddedV5ProfileTag shall be as defined in ISO 20677-1, with the specific requirements shown in Table 3.

Table 3. Header requirements

Header field	Required content
Profile class	'prn '
Profile subclass	'sref'
Profile subclass major version	1
Profile subclass minor version	0
Profile flags	Embedded
Device attributes	
Data colour space	Same as Data colour space in encapsulating ICC profile
MCS	0
Colorimetric PCS	0
Spectral PCS	'rsxxxx' where xxxx is a hex value corresponding to the number of spectral channels in the PCS
Spectral PCS range	Start and end wavelengths of the spectral PCS encoded as float16 values
Bispectral PCS	0

Full details of the encoding of the header fields in Table 3 are given in ISO 20677-1.

6.2.1 Required tags

Profiles shall contain the tags listed in Table 4.

Table 4. Required tags

Tag name	Signature	Required content
spectralWhitePointTag	'swpt'	float16NumberType or float32NumberType containing an array of values defining the spectral reflectance or emission of the media white point
DToB3	'DToB3'	multiProcessElementType element containing zero or more curveSetElement (excluding sampledCalculatorCurve), matrixElement, CLUTElement, extendedCLutElment processing elements
spectralViewingConditionsTag	'svcn'	Structure defining Standard 2-degree observer, and D50 illuminant

The encoding of the tags listed in Table 4 shall be as defined in ISO 20677-1.

6.2.2 Example

An example hybrid printer profile that is encoded according to the requirements of this ICS document is provided alongside this specification in the iccDEV-Testing ICS package named HybridPrinterWithReflectance. The example profile is built from CMYK_Hybrid_Profile.xml by the BuildAndTest.bat (Windows) and BuildAndTest.sh (POSIX) scripts that accompany the package. The package also provides supporting colorimetric, spectral and multispectral profiles, together with JSON driver files that exercise each of the workflow scenarios defined in clause 5.2.3 against this profile. Annex A summarises the worked example scenarios that are provided.

7 Conformance

A profile shall be considered to be in conformance with this ICS document if it meets the following conditions:

- The profile connects to the channels specified in section 5.
- The profile header includes the required content from Table 3.
- All required tags listed in Table 4 are present in the profile.
- The profile structure and all tags conform to ISO 20677-1.

A CMM shall be considered to be in conformance with this ICS if it meets the following conditions:

- The CMM is able to parse profiles that conform to this ICS

- The CMM supports and is capable of processing the channels specified in section 5 and any other profiles listed in Table 2.
- The CMM is able to process the tags listed in Table 4.
- When processing a profile conforming to this ICS, the CMM produces results that are a close approximation to those produced by the iccMAX Reference Implementation [3]

Bibliography

[2] iccMAX http://www.color.org/iccmax/

[3] iccMAX Reference Implementation http://www.color.org/iccmax/index.xalter#reficcmax

8 Annex A (informative) - Worked example scenarios

This annex describes the worked example scenarios provided in the iccDEV-Testing ICS package named HybridPrinterWithReflectance. Each entry is driven by a JSON configuration file in the package's config/ directory and demonstrates one of the workflow scenarios defined in clause 5.2.3 applied to the example hybrid printer profile.

Table A.1. Worked example scenarios in the iccDEV-Testing ICS package

ID	Clause 5.2.3 scenario	iccDEV tool / driver file	What is demonstrated
S1	Scenario 1 (5.2.3.2)	iccApplyProfiles / hpwr-S1-PrintOutput.json	Rendering an embedded-sRGB RGB image into CMYK through the hybrid printer profile using its base part (perceptual).
S2	Scenario 2 (5.2.3.3)	iccApplyProfiles / hpwr-S2-PrintProof.json	Colorimetric (D50, absolute) proof of the CMYK output back to sRGB using the base part of the hybrid printer profile.
S3	Scenario 3 (5.2.3.4)	iccApplyNamedCmm / hpwr-S3-SpectralPcsAccess.json	Direct access to spectral PCS reflectance values for

			two CMYK greys via a colour-list pipeline through the iccMAX sub-profile, with no spectral PCS processing performed.
S4a	Scenario 4 (5.2.3.5)	iccApplyProfiles / hpwr-S4a-SpectralPrintProof.json	Spectral proof under CIE Illuminant A using the iccMAX sub-profile of the hybrid printer profile with a PCC override.
S4b	Scenario 4 (5.2.3.5)	iccApplyProfiles / hpwr-S4b-SpectralPrintProof.json	Spectral proof under CIE Illuminant D93 using the iccMAX sub-profile of the hybrid printer profile with a PCC override.
S5a	Scenario 5 (5.2.3.6)	iccApplyProfiles / hpwr-S5a-SpectralExtraction.json	Extraction of a multispectral RGB image from the CMYK output via the iccMAX sub-profile and a multispectral RGB destination profile.
S5b	Scenario 5 (5.2.3.6)	iccApplyNamedCmm / hpwr-S5b-SpectralExtraction.json	Extraction of the reflectance spectrum for two CMYK greys via a colour-list pipeline through the iccMAX sub-profile.
S6	Scenario 6 (5.2.3.7)	iccApplySearch / hpwr-S6-SpectralReproduction.json	Inverse spectral reproduction: given the reflectances from S5b, search for CMYK values that reproduce them under D93, A, D50 and F11 with equal PCC weights.

S5b and S6 together form a spectral round trip: S5b produces reflectance spectra from CMYK input, and S6 searches for CMYK values that reproduce those spectra through the example hybrid printer profile. The closeness of the S6 output to the original S5b input is a measure of the invertibility of the example profile under the given PCC weights.